ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. There are 594 documents in the sci.med category~.

2. The purpose of frequent pattern mining is to identify frequently occurring item combinations in a dataset. When analyzing common word combinations in the sci.med category, using the entire dataset would affect the calculation of support. By filtering the data to a specific category, we change the basis for calculating support, which can influence the results of frequent pattern mining and make the discovered patterns more relevant to the research domain~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to clarify the concepts of frequent pattern mining and support, and to translate and improve the wording of my original answer.

External references (links):

https://hackmd.io/community/@kcORo2iwRlWcr73qnlkRGg/DataMiningCh6?utm_source=community&utm_medium=rec

https://ithelp.ithome.com.tw/articles/10187244

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

1. Dropping rows with missing values is generally more appropriate when the proportion of missing data is small and removing those rows is unlikely to introduce significant bias. However, in this case, 40% of the documents are missing category labels. Using `dropna()` would be an overly draconian

approach, as it would result in the loss of a substantial amount of potentially valuable text data and could introduce sampling bias, thereby affecting the reliability of subsequent data mining results~.

2. Since the missing values are category labels rather than the text content itself, we can replace the missing category_name values with a specific string, such as Unknown, instead of deleting the entire documents. This approach preserves the original text data, allowing it to be used for word frequency analysis, text vectorization, or other data mining methods that do not require category labels, while avoiding arbitrary assumptions about the missing categories~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review the feasibility of my proposed missing data handling method, improve the clarity of my original Chinese response, and translate and polish it into English.

External references (links):

https://ithelp.ithome.com.tw/articles/10392641

https://master.get.com.tw/it/column/detail.aspx?no=913982

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The category proportions in the sampled dataset do not perfectly match those in the original dataset, but they are relatively close~.

2. When a dataset is highly imbalanced, simple random sampling may result in too few samples from minority classes, or even exclude them entirely, making the sampled dataset less representative of those classes. This may affect subsequent model training, causing the model to favor majority classes and reducing its ability to accurately predict minority classes~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review my understanding of sampling risks and to translate and polish my original Chinese answers into English.

External references (links):

https://ithelp.ithome.com.tw/articles/10261413

https://vocus.cc/article/66d01b68fd897800013faca5

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features increases~.

2. Adding common stopwords back into the dataset generally increases the proportion of non-zero values in the document-term matrix, thereby reducing its overall sparsity~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to understand the concepts of text vectorization and matrix sparsity, review my original answers, and translate and polish them into English.

External references (links):

https://vocus.cc/article/683302c3fd897800015532f3

https://juejin.cn/post/7340898858555867136

https://ithelp.ithome.com.tw/m/articles/10296818

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

1. As the variance threshold increases from 0.0 to 0.05, the vocabulary size decreases~.

2. Words with near-zero variance typically include rare words that appear in very few documents and common words that occur with similar frequencies across almost all documents. Since their occurrence counts vary very little between documents, these words are usually removed first~.

3. Features with near-zero variance show very little variation across documents and generally provide limited information for distinguishing between them. Therefore, removing these features can reduce redundant information and noise, decrease dimensionality, and lower computational costs without losing much useful information~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review my understanding of variance-based feature selection, check the accuracy of my original Chinese answers, and translate and polish them into English.

External references (links):

https://www.taroballz.com/2018/07/06/ML_DecreaseFeature/

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Pearson's top word is "thanks" (r = 0.278), a general word, while Spearman's top word is "graphics" (r = 0.521), a domain-specific term~.

2. Fundamentally, Pearson measures linear correlation using raw values, while Spearman measures monotonic correlation using ranks and is less sensitive to outliers~.

3. When a word appears unusually frequently in a few documents, extreme counts may weaken its Pearson correlation, while Spearman can remain high because it uses ranks~.

AI used for this answer: Yes

If yes, how: If yes, how: I used ChatGPT to review the statistical accuracy of my original Chinese answers, clarify the differences between Pearson and Spearman correlation, and translate and polish my responses into English.

External references (links):

https://ithelp.ithome.com.tw/articles/10245037

https://www.youtube.com/watch?v=5lCLpNudTio

https://apxml.com/zh/courses/intro-eda-course/chapter-4-bivariate-analysis/numerical-vs-numerical-correlation

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup: When the minimum support threshold (minSup) in MaxFPGrowth decreases from 9 to 3, the total number of discovered maximal frequent patterns increases from 710 to 9,547, while the execution runtime increases from 0.3307 seconds to 6.6452 seconds~.

Answer (2) — Top-K k vs MaxFPGrowth minSup: Fundamentally, Top-K FAE ranks patterns based on their support and selects the top k patterns. Therefore, k controls the number of patterns to retain. In contrast, MaxFPGrowth uses minSup as the minimum support threshold and retains only maximal frequent patterns that meet this threshold and have no larger frequent supersets. Thus, minSup determines the minimum number of occurrences required for a pattern to be considered frequent, rather than directly specifying the number of patterns to output~.

Answer (3) — Maximal vs standard frequent patterns: Maximal frequent patterns are patterns that meet the minimum support threshold and have no larger frequent supersets. Since MaxFPGrowth does not need to list all the frequent subsets of these patterns, it reduces redundant information and produces a more concise representation than the complete set of frequent patterns~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review the accuracy of my original Chinese answers and translate and polish them into English. The experimental results were obtained by running the code myself.

External references (links):

https://www.youtube.com/watch?v=-ao-K70q1D4

https://ctld.video.nccu.edu.tw/media/6863

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. The original X_counts matrix contains 35,482 word features, meaning that each document is represented in a high-dimensional feature space. However, a two-dimensional scatter plot has only two coordinate axes, so it cannot directly represent all the original features without losing information~.

2. Fundamentally, dimensionality reduction algorithms, such as PCA, t-SNE, or UMAP, must be applied first to transform the original 35,482-dimensional data into a two-dimensional representation. This allows each document to be plotted using two coordinates, making it possible to visualize the distribution and clustering of different document categories~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review the accuracy of my original Chinese answers and translate and polish them into English.

External references (links):

https://ithelp.ithome.com.tw/m/articles/10267685

https://www.pershing.com.tw/ai-%E5%85%A5%E9%96%80%EF%BC%9A11-%E8%B3%87%E6%96%99%E6%BA%96%E5%82%99%EF%BC%8D%E7%B6%AD%E5%BA%A6%E7%B8%AE%E6%B8%9B/

https://medium.com/d-d-mag/%E6%B7%BA%E8%AB%87%E5%85%A9%E7%A8%AE%E9%99%8D%E7%B6%AD%E6%96%B9%E6%B3%95-pca-%E8%88%87-t-sne-d4254916925b

https://hackmd.io/@YungHuiHsu/rk-G8BpBT

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP)** shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. In these visualizations, UMAP shows clearer regional differences between document categories compared to PCA and t-SNE. Some documents form relatively distinct clusters, although the four categories still overlap to some extent. In contrast, most data points in the PCA plot are concentrated near the origin, with substantial overlap between categories. While t-SNE reveals some local clusters, many data points from different categories remain mixed in the main central region~.

2. PCA is a linear dimensionality reduction technique that projects high-dimensional data into a lower-dimensional space by identifying directions of maximum variance. However, the directions with the greatest variance are not necessarily the most effective for separating document categories. In addition, high-dimensional text data may contain complex nonlinear relationships that PCA cannot effectively capture. In contrast, t-SNE and UMAP are nonlinear dimensionality reduction techniques that focus more on preserving neighborhood relationships between data points, making it easier to reveal local clusters of different document categories in two-dimensional scatter plots~.

3. t-SNE is a nonlinear dimensionality reduction technique that aims to preserve local neighborhood relationships between data points in high-dimensional space rather than accurately preserving global distances between separate clusters~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to translate my original Chinese answers into English and improve their clarity and academic wording.

External references (links):

https://www.youtube.com/watch?v=cHC8gQ1Odjk

https://morrishare.com/three-dimensionality-reduction-method-comparison/

https://ithelp.ithome.com.tw/m/articles/10267685

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?

3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four binary columns are created to represent the four categories~.

2. If integer category labels are directly used as input features, the algorithm may incorrectly assume that the categories have a meaningful order and numerical distances between them. However, these integers are simply labels used to identify different categories~.

3. Fundamentally, one-hot binarization represents each category using a separate binary position, where the position corresponding to the category is set to 1 and all other positions are set to 0. Therefore, the algorithm no longer interprets the categories based on the numerical order or differences between their original integer labels, thereby avoiding false assumptions about ordering and distance~.

AI used for this answer: Yes

If yes, how: I used ChatGPT to review the accuracy of my original Chinese answers and translate and polish them into English.

External references (links):

https://samriddhl.medium.com/binary-encoding-vs-one-hot-encoding-choosing-the-right-approach-for-your-ml-models-9be4533e8772
